Bharadwaj Manne

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SUMMARY

Graduate Computer Science student at Florida State University with strong experience in developing backend and distributed systems through academic research and industry internships. Strong background in Java, Python, and C++ with solid foundations in data structures, algorithms, and object oriented design. Designed and deployed cloud based services on AWS, GCP and Azure using Docker, Kubernetes, and CI/CD pipelines. Experience includes debugging complex systems, analyzing performance behavior, and improving reliability and scalability through profiling, testing, and careful system design.

EDUCATION

Florida State University, Tallahassee, Florida — Aug 2024 – May 2026
Master of Science in Computer Science — **GPA:** 3.84 / 4.0

Vellore Institute of Technology, Andhra Pradesh, India — Aug 2020 – May 2024
Bachelor of Technology in Computer Science and Engineering (Data Analytics) — **CGPA:** 9.38 / 10.0

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C++, C, JavaScript, SQL, Bash, R
- **Web and API Development:** HTML, CSS, React, Node.js, Spring Boot, Flask, REST APIs, JSON, XML
- **Databases and Data Modelling:** MySQL, PostgreSQL, MongoDB, DynamoDB, ETL, Snowflake, Star Schema
- **Cloud Platforms:** Amazon Web Services (AWS), Google Cloud Platform, Microsoft Azure, Chameleon Cloud
- **DevOps and Systems:** Git, Docker, Kubernetes, Terraform, Jenkins, Apache Airflow, Ubuntu, Shell Scripting
- **Testing and Quality Assurance:** JUnit, PyTest, Postman API testing, A/B testing
- **Machine Learning:** Scikit Learn, TensorFlow, PyTorch, OpenCV, Pandas, NumPy
- **Data Analytics and Reporting:** Microsoft Excel, Power BI, Tableau, SQL Server Reporting
- **Business and Enterprise Tools:** Salesforce CRM, ServiceNow, Jira Ticketing, Microsoft Power Automate
- **Development Tools:** VS Code, IntelliJ IDEA, PyCharm, Jupyter Notebook, Eclipse, GDB, Make, CMake

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, Florida State University, Tallahassee, FL — Aug 2025 – Present

- Supported operating systems coursework for 300+ students by conducting recitation sessions, leading lab activities, and reinforcing systems programming concepts using C, C++, Python, and Rust.
- Managed and maintained Proxmox based virtualization environments, supporting process isolation, file system concepts, and virtual machine provisioning for operating systems assignments and projects.
- Held regular office hours and evaluated assignments and labs while maintaining academic and grading records using Canvas, Microsoft Excel, and Power BI.

Graduate Research Assistant, Florida State University, Tallahassee, FL — May 2025 – Present

- Conducted research on concurrency bugs, crash consistency, and persistent memory correctness in PM based file systems as part of the Silhouette bug detection framework under the supervision of Dr. An-I Wang.
- Developed LLVM and Clang based instrumentation in C++ to trace synchronization events, memory operations, and vector clocks for race detection and analysis in PM file systems such as PMFS, NOVA, and WineFS.
- Built multithreaded tracing, and benchmarking workflows, reducing bug exploration space by 10x, enabling up to 8x faster bug discovery, and achieving up to 28% throughput improvement through profiling and monitoring.

Cloud Computing Intern, Unical Systems, Hyderabad, India — Dec 2023 – Apr 2024

- Deployed and supported AWS hosted microservices by provisioning and maintaining the EC2 environments, managing S3 workflows, IAM access controls, and monitoring system health and logs through CloudWatch.
- Supported backend REST service development and integrations using Java and MySQL, validated APIs using Postman, and assisted with data access workflows involving DynamoDB.

- Automated infrastructure provisioning and CI/CD pipelines using Terraform, Jenkins, Docker, and Kubernetes, improving deployment consistency and contributing to a 39% reduction in deployment downtime.

Applied Data Science Extern, SmartInternz, Remote — May 2023 – Jul 2023

- Built an end to end analytics and modeling workflow on a 3,500+ record dataset using Python, Pandas, NumPy, and SQL by cleaning missing and duplicate data and preparing train and test splits for supervised learning.
- Trained and tuned scikit learn models using cross validation, achieving 96% accuracy, and deployed results through a Flask REST API and Power BI dashboard.

Java Development Intern, Oasis Infobyte, Remote — Sep 2022 – Oct 2022

- Designed a payroll management system in Java (Spring Boot + MySQL), improving efficiency by 47%.
- Implemented secure authentication and RESTful APIs for data management and transactional workflows.
- Performed debugging, unit testing, and code optimization to improve system reliability and performance.

PROJECTS

NIFTY – Decentralized Fault Tolerance Framework for Distributed Systems — **GitHub:** Bharadwaj-1953/NIFTY

- Built a decentralized fault monitoring framework in C++23 using heartbeat based failure detection and traffic monitoring to handle partial network partitions, eliminating central dependency and improving scalability.
- Evaluated the system in multi node environments, achieving up to 99.5% fault detection accuracy with <3% CPU and network overhead under simulated failure conditions.

MDUAL – Multiple Dynamic Outlier Detection for Streaming Data — **GitHub:** Bharadwaj-1953/MDUAL

- Designed and implemented a real time outlier detection algorithm for streaming data using data and query duality to reduce redundant distance computations.
- Implemented the system in C++17 and demonstrated upto 221× runtime improvement while reducing memory usage by 13× compared to existing approaches such as SOP and pMCSKY.

AWS MedData – Cloud Based Medical Data Integration Platform — **GitHub:** Bharadwaj-1953/AWS-MedData

- Developed a cloud based healthcare data integration application using Python and Flask, deployed on AWS EC2 with secure storage in Amazon S3.
- Implemented Locality Sensitive Hashing (LSH) to support privacy aware querying and designed a modular architecture for Admin, Doctor, Patient, and IOH users backed by a PostgreSQL database.

PUBLICATIONS & PATENTS

Systems and Methods for Detection of Anomalies in Civil Infrastructure Using Context-Aware Semantic Computer Vision Techniques

Publisher: IPO | Application Number: 202321049964

- Proposed a context-aware anomaly detection framework using CNNs for semantic visual feature extraction with rule-driven contextual reasoning to detect structural defects, and abnormal patterns in video streams.
- Integrated spatiotemporal context modeling and configurable rule thresholds to correlate visual semantics across frames, reducing false positives by 27% and enabling reliable real-time anomaly detection.

Dilated Convolutions and Time-Frequency Attention for Speech Enhancement

Publisher: IEEE | Accession Number: 23204218

- Developed a Dilated Time Frequency Attention Autoencoder (DTFAAEC) using CNNs to enhance noisy speech by modeling long range temporal and spectral patterns.
- Demonstrated a 34% improvement in speech intelligibility over baseline models on STOI and PESQ metrics using Python and TensorFlow while maintaining low latency inference for real time audio processing.

CERTIFICATIONS

- **Google Cloud Certified:** Associate Cloud Engineer
- **AWS Academy Graduate:** Cloud Foundations
- **Microsoft Certified:** Azure AI Fundamentals (AI-900), Azure Data Fundamentals (DP-900), Power Platform Fundamentals (PL-900), Azure Fundamentals (AZ-900)
- **Salesforce Certified:** AI Associate, Agentforce Specialist
- **Cisco Certified:** Cybersecurity Essentials, Introduction to IoT